Boundaries and 'Not Sure'

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

TEACHER GUIDE · 1 OF 9

Boundaries and 'Not Sure'

> Prepare Boundaries and 'Not Sure': check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Identify a specific source of **uncertainty** and explain how it affects a **prediction**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Save New Examples for Testing

Without opening the old lesson, explain one key idea from Save New Examples for Testing. Give an example and a mistake that the idea helps you avoid.

Look for: Separate examples used to choose a model from examples reserved for its final evaluation

Reference example, not the only acceptable wording: Comparing candidates is development. Scoring a frozen model on previously reserved examples is evaluation, provided they did not influence its development.

Changing a threshold after reading the test score uses that test information for development of the revision; the same set is no longer independent of that revision.

Learning from Examples

Without opening the old lesson, explain one key idea from Learning from Examples. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a threshold by tracing **prediction**s and counting disagreements for every candidate

Reference example, not the only acceptable wording: Threshold 3 predicts short, tall, tall, tall, disagreeing only at 4 cm: count 1. Threshold 7 predicts short, short, short, tall, disagreeing only at 6 cm: count 1.

Reset the tally for each candidate.

If/Then Decisions

Without opening the old lesson, explain one key idea from If/Then Decisions. Give an example and a mistake that the idea helps you avoid.

Look for: Evaluate a condition using the current state and trace only the selected branch

Reference example, not the only acceptable wording: Start 2: after adding 2 score is 4; at least 6 is false; add 4 and report 8. Start 5: after adding 2 score is 7; at least 6 is true; subtract 3 and report 4.

Each run starts fresh and executes only its selected branch.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A sorter receives a clear purple card, a red card hidden by a sleeve and a card labelled both red and blue by two reviewers. Give a different reason for **uncertainty** in each case and a suitable next action.

Concept to explain

Overlapping categories create **uncertainty**; **abstain**ing can be responsible.

Term	Meaning
boundary case	example near competing category rules
uncertainty	limited support for one conclusion
abstain	decline to choose a category

Term	Meaning
escalation	send the case for added information or review

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Develop **uncertainty** as a specific **evidence** gap with a matching review path. The possible-**input** comparison is a reasoning exercise, not a probability calculation.

Assume the listed possibilities are complete for the paper tasks; do not imply that a model normally knows all **possible input**s or that shared **prediction**s verify correctness.

Readiness answer

At 5 it predicts short; at 6 it predicts tall. The unclear **input** leaves two different possible **prediction**s. A previous test result does not tell us which height this record contains.

Tower cards labelled either 5 or 6 cm, either 8 or 9 cm, and height missing

A review record with original **input**, model version, issue, requested **evidence** and outcome

Distance possibility cards from W4; use fictional measurements and no personal data

End goal for the pictorial case

By the end, I can use **uncertainty** to request a specific review when **possible input**s lead to different decisions.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Boundaries and 'Not Sure' using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT features with incomplete or conflicting clues

2. Process — the deciding step

PROCESS compare **evidence** with category boundaries

3. Output — the result

OUTPUT category, not sure, or request **more information**

Key idea

CORE IDEA Overlapping categories create **uncertainty**; **abstain**ing can be responsible.

How the mechanism works

UNDER THE HOOD Abstention is a decision policy: when confidence or **evidence** fails a threshold, the system defers.

Good thresholds balance coverage against error cost and specify what additional information or human review follows.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE A soft toy shaped like a ball may fit toy, sports, or comfort-item bins depending on the sorter's purpose.

Step 1

STEP 1 — OBSERVE features with incomplete or conflicting clues

Step 2

STEP 2 — TRACE compare **evidence** with category boundaries

Step 3

STEP 3 — CONCLUDE category, not sure, or request **more information**

Model answer

MODEL REASONING A useful not-sure result is specific: it states which categories compete, what **evidence** is missing, and who or what should supply it.

Abstention reduces forced errors but should not become an excuse to avoid all decisions.

A claim to avoid

DO NOT OVERCLAIM Responsible **uncertainty** is informative and actionable, not a blank refusal.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

A sorter handles clearly red and clearly blue cards. Purple and very dark cards arrive.

How should it handle each uncertain case?

Exact explanation to model

Why the result follows

Use a review tray. Purple is outside the clear categories; darkness prevents **observation**. These are different **uncertainty** causes.

Explain the added worked example

Explain the added worked example · 1

An unclear measurement, a missing feature and a weakly tested model create different problems. Saying the AI is unsure does not identify which one matters.

Write what is known, what remains unknown and how the unknown could change the result. Then choose a check that addresses that specific gap.

Use the familiar tower model: tall at a height of at least 6 cm, short otherwise, for whole-number heights from 1 through 10. A smudged card could read 5 cm or 6 cm. The threshold is known; the **input** is unclear.

Repeating the same calculation cannot recover the missing digit.

Explain the added worked example · 2

For this paper investigation, treat the listed possibilities as the complete set allowed by the exercise. In real use, that assumption would itself need **evidence**.

We will compare the model's **prediction**s under each possibility; we are not changing Tower1 to accept an invented exact measurement.

Explain the added worked example · 3

At either 5 or 6 cm, one possibility predicts short and the other tall. Record no single category yet and ask for the original record or a new measurement. Do not select the more convenient digit.

Averaging the two readings would introduce a different **input** that neither possibility stated.

Explain the added worked example · 4

At either 8 or 9 cm, the model predicts tall in both cases. You can state that the **prediction** stays tall under these two possibilities. Keep the exact-height **uncertainty** visible.

This agreement does not check the model against a correct task label or establish that the list of possible heights is complete.

Key terms

Term	Meaning
Uncertainty	Something relevant that the available evidence does not settle

Term	Meaning
Possible input	An input still compatible with the stated information
Review trigger	A stated condition that sends a case for an additional check
Review evidence	Information gathered to investigate the reason a case was referred
Unresolved	Still not settled after the available checks

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Reason for review	Relevant check	What it cannot settle alone
Smudged height	Read the original or remeasure	Whether the model is accurate
Unit missing	Check how the record was made	The unit from digits alone
Two labels for one record	Check the label policy and sources	Truth by majority vote
Poorly tested new use	Evaluate relevant reserved cases	Every future outcome

Model the reasoning aloud · 5

A review request should carry the original **input**, model version, **unresolved** issue and any **prediction**s already made. Name the **evidence** to seek and the person responsible for checking it.

A reviewer needs access to relevant **evidence** and a clear task; being human does not make an unsupported guess correct.

Model the reasoning aloud · 6

After review, record what was checked and what changed. If a reliable new reading establishes 6 cm, the model predicts tall. If the record is still unreadable, keep no **prediction** with reason **unresolved** height.

Do not replace the original record without retaining the reason for the correction.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Boundaries and 'Not Sure'

A fictional cubby selector accepts a block if width is at most 5 cm.

The facts we will use

Supplied facts

An unclear measurement means Block A is either 4 or 5 cm; Block B is either 5 or 6 cm.

Supplied facts

These complete possibility sets contain all widths under consideration. Other dimensions are known to fit.

AI ACADEMY / AI-1/16

Possibilities and decisions

	Width 1 cm	Decision 1	Width 2 cm	Decision 2
Block A	4	fits	5	fits
Block B	5	fits	6	fails

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/16

Boundaries and 'Not Sure'

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Block A possibilities

4 and 5 both satisfy the width limit, which includes equality.



2. Block B possibilities

5 fits; 6 does not.



3. Compare decisions

A has one supported decision; B's possible decisions disagree.



4. Review B

Ask a named reviewer to remeasure B using the specified width unit.

CHANGE ONE THING

Narrow B's possibilities to 6 or 7 cm.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Block A possibilities	4 and 5 both satisfy the width limit, which includes equality.
2. Block B possibilities	5 fits; 6 does not.
3. Compare decisions	A has one supported decision; B's possible decisions disagree.
4. Review B	Ask a named reviewer to remeasure B using the specified width unit.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Uncertainty about a number need not always create **uncertainty** about the decision. Every allowed width for A fits, so its width **uncertainty** does not affect acceptance. B straddles the boundary.

Record both possibilities and request a better measurement instead of selecting the one that makes a preferred outcome.

What this example does not establish

Keep the stated limit

A review path is not automatically accurate; its conclusion depends on relevant, better **evidence**.

Change one thing in the pictured case

Change one thing

Change: Narrow B's possibilities to 6 or 7 cm.

Predicted result

Prediction: Both fail the width limit; rejection no longer depends on resolving their difference.

Why the result follows

Reason: All remaining possibilities lead to the same decision.

Ask while showing the picture

Ask while showing the picture · 7

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Boundaries and 'Not Sure', reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A sorter handles clearly red and clearly blue cards. Purple and very dark cards arrive.

2. State the decision to explain

Write the question to explain: How should it handle each uncertain case?

3. Reveal the worked reasoning

Reveal after **prediction**s: Use a review tray. Purple is outside the clear categories; darkness prevents **observation**. These are different **uncertainty** causes.

Hold back the independent task

Keep for an independent attempt: A sorter knows pencils and books. You show a lunch box. Should it force a known label or ask for help? State a useful fallback.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Record the original 5-or-6 cm card, threshold-6 model version, possible **prediction**s short and tall, and **unresolved** height. Request a reliable original reading or remeasurement.

A supported 6 cm reading permits a tall **prediction**, with the correction documented. If no reliable value is obtained, retain no **prediction** and hold sorting under the policy.

Exit: repeated runs show consistent computation on the guessed **input**; a reliable height is still needed.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve the two threshold outcomes for 5 and 6. Distinguish an unclear input from a known model rule.
7 to 20 minutes	Trace every diagram row. State the complete-possibilities assumption and retain exact-height uncertainty even when predictions agree.
20 to 33 minutes	Complete W1 and W2. Ask students to identify which claim follows from both possible inputs and which claims still lack evidence.

Stage	Teacher action
33 to 45 minutes	Use W3 and the review table to match each missing fact to a useful source. Reject an assumed unit as a repair.
Session 2 start 0 to 8 minutes	Retrieve the difference between a new measurement and another run on an unchanged input.
8 to 23 minutes	Construct the guided review record and both endings. Require the unresolved path to stop without inventing a height.
23 to 38 minutes	Collect W4 independently. Look for all three possibility comparisons, an appropriate measurement check and limited claims.
38 to 45 minutes	Discuss W5 and the exit. A reviewer can preserve an unknown; human involvement alone is not correctness evidence.

Reduce help deliberately

Completion practice

Completion task: Assume the only possible heights are 8 and 9 cm. Apply tall at least 6 to each. Write a justified result statement and two things it does not establish. Do not invent a confidence percentage.

Reasoning to listen for

Teacher check: Both **input**s predict tall, so the **prediction** is tall under both listed possibilities. This does not identify whether the exact height is 8 or 9 and does not establish agreement with the correct task label.

Outside the exercise, completeness of the possibilities also needs **evidence**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Uncertainty** and review. Proposed explanation: “Two possible values imply equal probabilities”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: A list of possibilities supplies no probabilities unless further **evidence** states them.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Which original block needs a better width before accepting or rejecting it? What should the reviewer do?

Reasoned teacher answer

B needs review because 5 fits and 6 fails. The reviewer should remeasure width in cm and retain the **evidence**.

Changed-case reasoning: Both fail the width limit; rejection no longer depends on resolving their difference. All remaining possibilities lead to the same decision.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Boundaries and 'Not Sure' investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Find three **boundary case**s and design a 'need more info' **output**.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

Find the unsupported claim

CLAIM TO REPAIR Not sure always means the system has failed.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Two possible values imply equal probabilities: A list of possibilities supplies no probabilities unless further **evidence** states them.

Averaging guesses recovers the missing **input**: A calculated average is a new assumption, not an observed measurement.

Repeated identical **output**s remove **input uncertainty**: Re-running unchanged information does not reveal a missing digit or unit.

A human reviewer can always settle the case: Require relevant **evidence** and allow an explicit **unresolved** outcome.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Boundaries and 'Not Sure' questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Use a review tray. Purple is outside the clear categories; darkness prevents **observation**. These are different **uncertainty** causes.

Independent question

Independent question · withhold the key

A sorter knows pencils and books. You show a lunch box. Should it force a known label or ask for help? State a useful fallback.

Expected independent answer

Teacher key · Expected independent answer

It should flag an unfamiliar item and ask a person. A useful fallback is not sure, please review. A known label would overstate the available **evidence**.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions

Criterion	0 points	1 point	2 points
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Not sure always means the system has failed.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Purple needs a policy for an uncovered or boundary category; remove the sleeve or obtain a usable **observation** for the hidden card; inspect definitions and adjudicate disagreement for conflicting labels.

A review outcome is useful only when it leads to an appropriate action.

Explanations for the additional workbook examples

W1 Describe the uncertainty precisely

Workbook question · W1

For a smudged height that could be 5 or 6 cm, state what is known, what is unknown and the two **prediction**s under tall at least 6. Choose a check that could distinguish the heights.

Explain why rerunning the threshold comparison cannot do that.

Teacher reasoning · W1

The rule and two stated possibilities are known; the actual height is **unresolved**. At 5 predict short, at 6 predict tall. Inspect the original readable record or remeasure.

Repeating a calculation supplies no new information about which height was recorded.

W2 Keep a stable prediction within its limits

Workbook question · W2

Assume the only possible heights are 8 and 9 cm. Apply tall at least 6 to each. Write a justified result statement and two things it does not establish. Do not invent a confidence percentage.

Teacher reasoning · W2

Both **input**s predict tall, so the **prediction** is tall under both listed possibilities. This does not identify whether the exact height is 8 or 9 and does not establish agreement with the correct task label.

Outside the exercise, completeness of the possibilities also needs **evidence**.

W3 Choose useful review evidence

Workbook question · W3

A record says 60 but omits the unit. A reviewer suggests reading it as 60 cm because the model needs cm. Explain the problem, name the **evidence** to request and state the outcome if the unit cannot be established.

Teacher reasoning · W3

The required **input** format does not prove the original measurement's unit. Ask for the original measurement record, instrument scale or recording instructions.

If the unit cannot be established, retain no **prediction** with reason **unresolved** unit; do not silently assume cm or convert from an assumed unit.

W4 Transfer the method to distance

Workbook question · W4

A paper model predicts far at a distance of at least 4 units, otherwise near, for whole-number distances 1 through 8. For each complete possibility set, compare **prediction**s: 2 or 3; 3 or 4; 4 or 5.

Identify the set requiring review before a single category can be stated. Design a relevant check and an **unresolved** ending. Limit the claims for the other two sets.

Teacher reasoning · W4

At 2 or 3 both predict near. At 3 predict near and at 4 far, so that set leaves the category **unresolved**. At 4 or 5 both predict far.

Request a reliable distance reading in the stated units for the disputed set; if unavailable retain no single category and reason **unresolved** distance.

The other two **prediction**s are stable only across their listed possibilities, not verified as correct labels or exact distances.

W5 Review the reviewer's conclusion

Workbook question · W5

A reviewer cannot read the original tower card but chooses 6 cm because two models both predict tall when given 6. Explain why this fails to resolve the **input**. Write a more defensible review record and a next step.

Teacher reasoning · W5

Both models were given the same guessed **input**, so their agreement does not establish the actual measurement.

Record that the original remains unreadable, the height is **unresolved** and no supported single-category **output** is available. Request a reliable new measurement or keep the sorting decision on hold under the stated policy.

Assess independent transfer

Independent question · withhold the key

Selected case: A paper model predicts far at a distance of at least 4 units, otherwise near, for whole-number distances 1 through 8. For each complete possibility set, compare **prediction**s: 2 or 3; 3 or 4; 4 or 5.

Identify the set requiring review before a single category can be stated. Design a relevant check and an **unresolved** ending. Limit the claims for the other two sets.

Selected case answer

Teacher key · Selected case answer

At 2 or 3 both predict near. At 3 predict near and at 4 far, so that set leaves the category **unresolved**. At 4 or 5 both predict far.

Request a reliable distance reading in the stated units for the disputed set; if unavailable retain no single category and reason **unresolved** distance.

The other two **prediction**s are stable only across their listed possibilities, not verified as correct labels or exact distances.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Boundaries and 'Not Sure' to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

Add a not-sure branch, missing-information prompt, and human review destination to the sorter.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 17: **Prediction** Is Not Fact. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use a number line and a separate card for each possibility. Keep known, unknown and requested **evidence** in three columns.

Extension: add a third possibility that changes an apparently stable result and explain why completeness of the possibility list matters.

W4 is secure with near/near, near/far and far/far comparisons, review of the middle set, useful **evidence** and an **unresolved** ending.

After support retry long at least 7 for whole cm lengths 1 through 10 using 4 or 5, 6 or 7, and 8 or 9: short/short, short/long, long/long. Require limits as well as labels.

Week 17 separates a **prediction** from the later **observation** or comparison **evidence** used to judge it, preserving both records.

Plan the delayed return

Suggested return points in this level: ai-1/17, ai-1/19, ai-1/22. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Add explicit review paths to the sorter, including **unresolved input**, responsible person and **evidence** needed.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Boundaries and 'Not Sure' lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Identify a specific source of **uncertainty** and explain how it affects a **prediction**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: Assume the only possible heights are 8 and 9 cm. Apply tall at least 6 to each. Write a justified result statement and two things it does not establish.

Do not invent a confidence percentage.

Supported practice answer

Reasoned practice answer: Both **input**s predict tall, so the **prediction** is tall under both listed possibilities.

This does not identify whether the exact height is 8 or 9 and does not establish agreement with the correct task label. Outside the exercise, completeness of the possibilities also needs **evidence**.

Check every learner before discussion: Use the worked case for **Uncertainty** and review. Proposed explanation: “Two possible values imply equal probabilities”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: A list of possibilities supplies no probabilities unless further **evidence** states them.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the

reasoning goal.

Extend the reasoning

Extension: Narrow B's possibilities to 6 or 7 cm. Require a **prediction** and a reason before comparing. Expected result: Both fail the width limit; rejection no longer depends on resolving their difference.

Reason: All remaining possibilities lead to the same decision.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

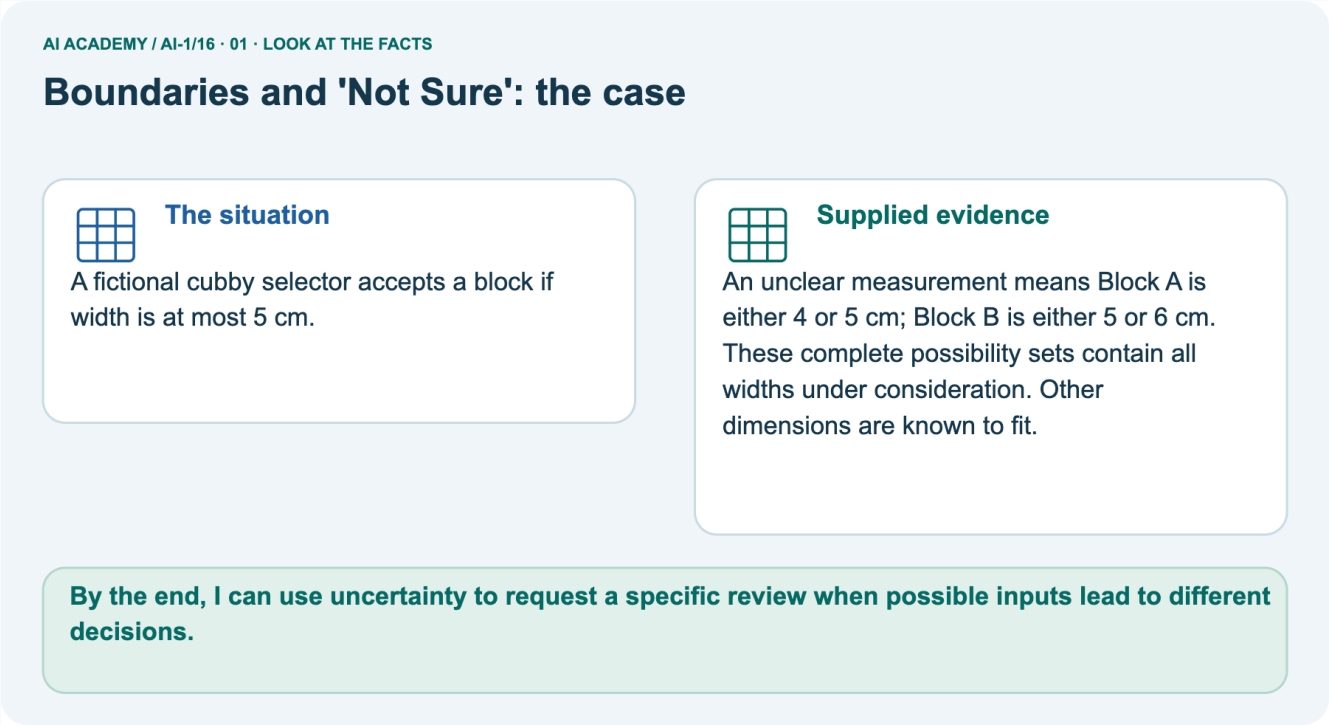
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Boundaries and 'Not Sure' graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Boundaries and 'Not Sure': the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional cubby selector accepts a block if width is at most 5 cm. Supplied facts: An unclear measurement means Block A is either 4 or 5 cm; Block B is either 5 or 6 cm.

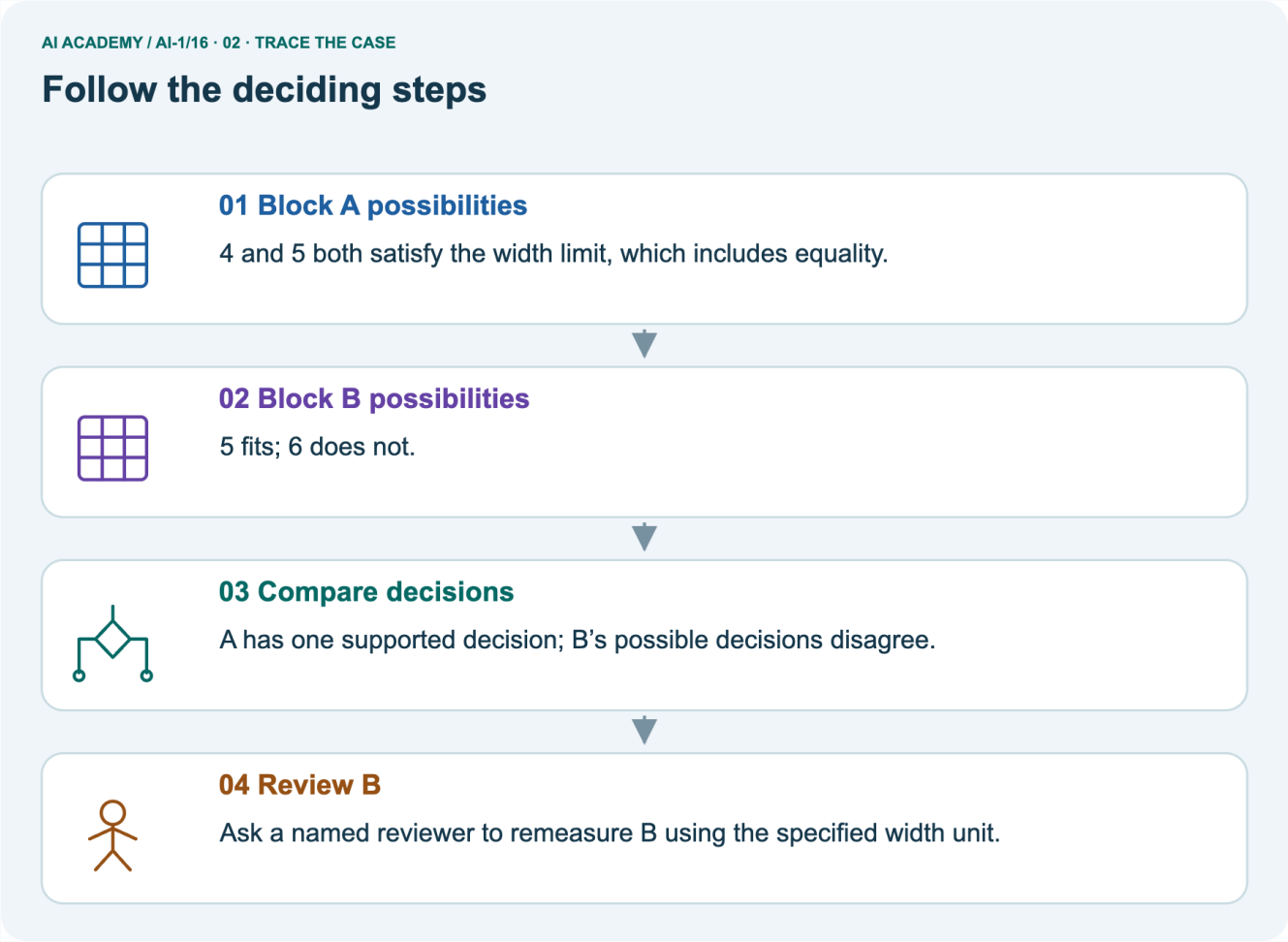
These complete possibility sets contain all widths under consideration. Other dimensions are known to fit.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

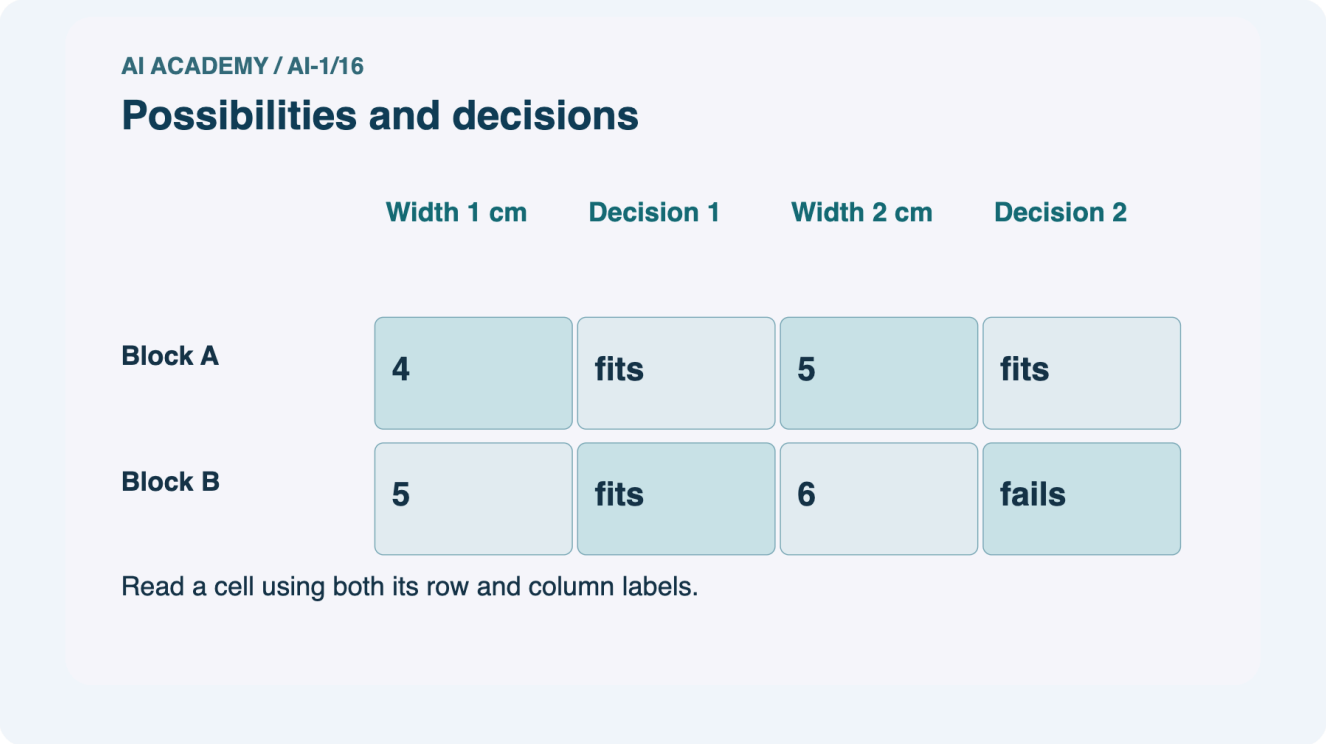
Block A possibilities: 4 and 5 both satisfy the width limit, which includes equality. → Block B possibilities: 5 fits; 6 does not. → Compare decisions: A has one supported decision; B's possible decisions disagree. → Review B: Ask a named reviewer to remeasure B using the specified width unit.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Possibilities and decisions



Follow the labelled picture with the matching explanation. Open the vector diagram

Possibilities and decisions. Block A: Width 1 cm = 4, Decision 1 = fits, Width 2 cm = 5, Decision 2 = fits. Block B: Width 1 cm = 5, Decision 1 = fits, Width 2 cm = 6, Decision 2 = fails.

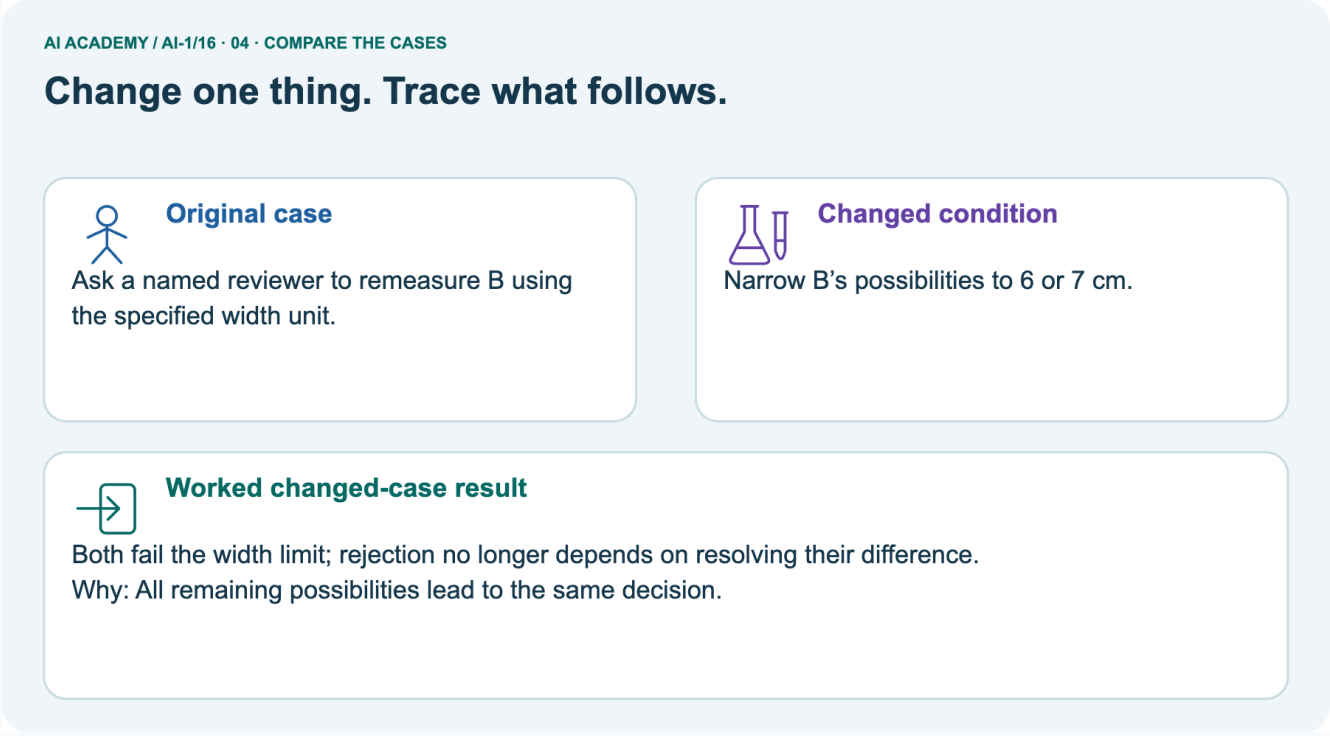
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Ask a named reviewer to remeasure B using the specified width unit. Changed condition: Narrow B's possibilities to 6 or 7 cm.

Expected result: Both fail the width limit; rejection no longer depends on resolving their difference. Reason: All remaining possibilities lead to the same decision.

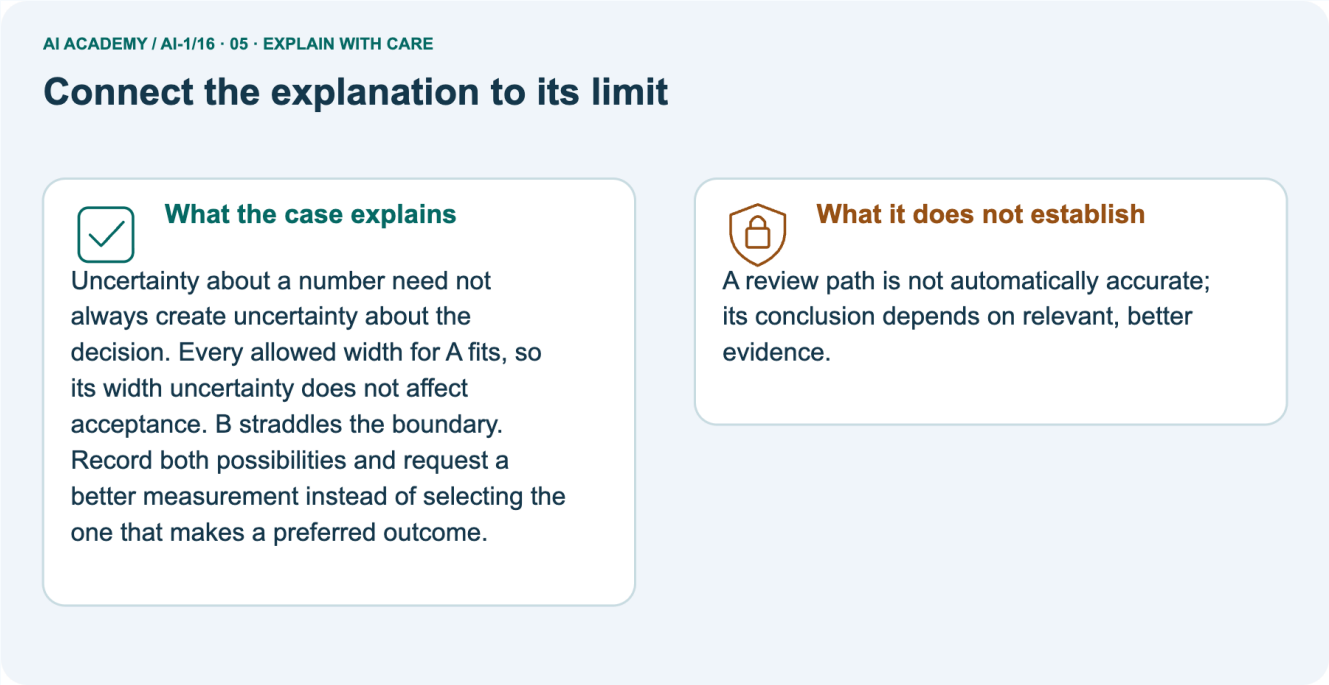
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: All remaining possibilities lead to the same decision.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Uncertainty about a number need not always create **uncertainty** about the decision. Every allowed width for A fits, so its width **uncertainty** does not affect acceptance. B straddles the boundary.

Record both possibilities and request a better measurement instead of selecting the one that makes a preferred outcome. Limit: A review path is not automatically accurate; its conclusion depends on relevant, better **evidence**.

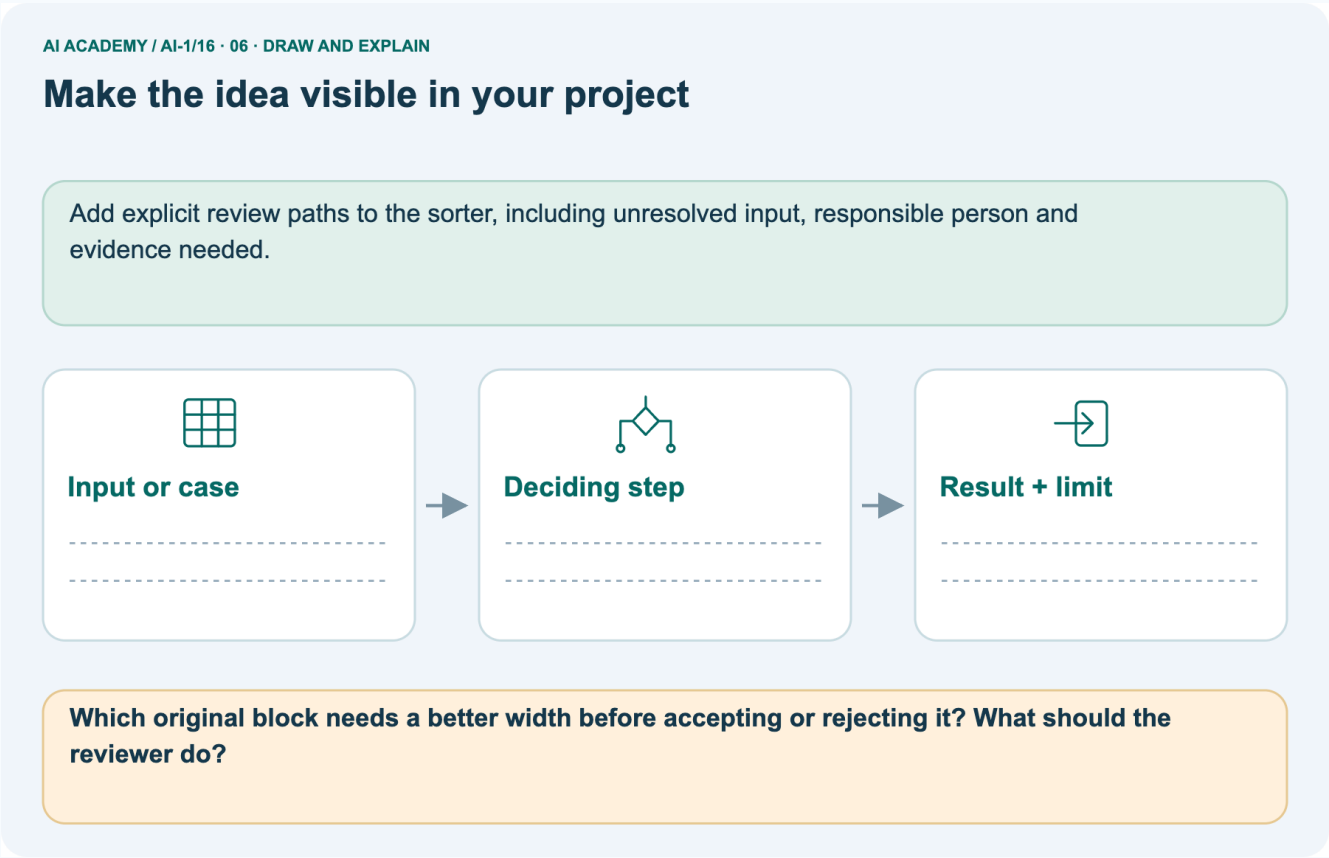
> Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Keep the stated scope: A review path is not automatically accurate; its conclusion depends on relevant, better **evidence**.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Add explicit review paths to the sorter, including **unresolved input**, responsible person and **evidence** needed. Draw your **input**, deciding step, result and limit.

Supported question: Which original block needs a better width before accepting or rejecting it? What should the reviewer do?

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: B needs review because 5 fits and 6 fails. The reviewer should remeasure width in cm and retain the **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.